



1. Description

1.1. Project

Project Name	HIL_2026
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	05/29/2026

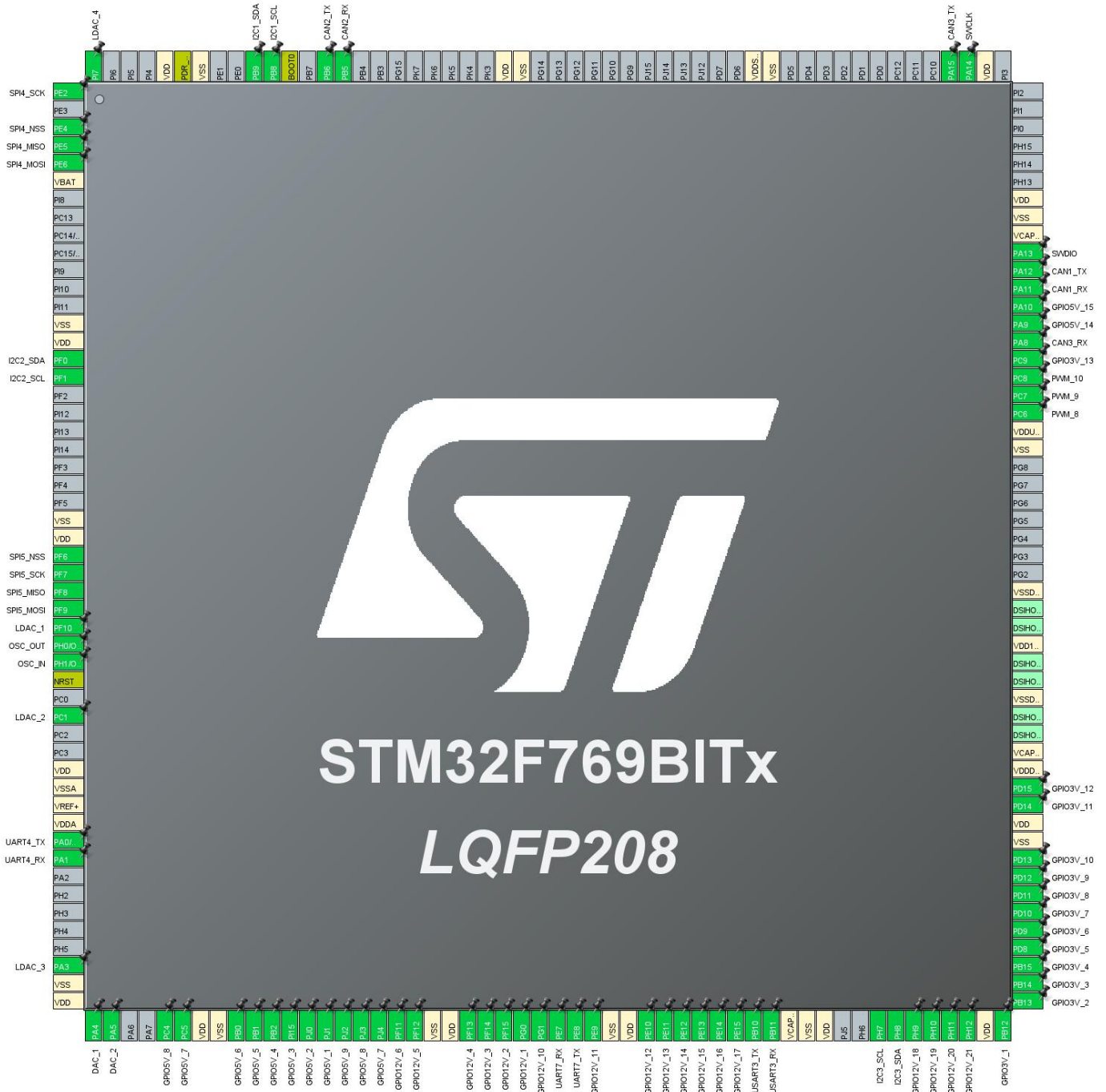
1.2. MCU

MCU Series	STM32F7
MCU Line	STM32F7x9
MCU name	STM32F769BITx
MCU Package	LQFP208
MCU Pin number	208

1.3. Core(s) information

Core(s)	Arm Cortex-M7
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	SPI4_SCK	
3	PE4	I/O	SPI4_NSS	
4	PE5	I/O	SPI4_MISO	
5	PE6	I/O	SPI4_MOSI	
6	VBAT	Power		
14	VSS	Power		
15	VDD	Power		
16	PF0	I/O	I2C2_SDA	
17	PF1	I/O	I2C2_SCL	
25	VSS	Power		
26	VDD	Power		
27	PF6	I/O	SPI5_NSS	
28	PF7	I/O	SPI5_SCK	
29	PF8	I/O	SPI5_MISO	
30	PF9	I/O	SPI5_MOSI	
31	PF10 *	I/O	GPIO_Output	LDAC_1
32	PH0/OSC_IN	I/O	RCC_OSC_IN	OSC_OUT
33	PH1/OSC_OUT	I/O	RCC_OSC_OUT	OSC_IN
34	NRST	Reset		
36	PC1 *	I/O	GPIO_Output	LDAC_2
39	VDD	Power		
40	VSSA	Power		
41	VREF+	Power		
42	VDDA	Power		
43	PA0/WKUP	I/O	UART4_TX	
44	PA1	I/O	UART4_RX	
50	PA3 *	I/O	GPIO_Output	LDAC_3
51	VSS	Power		
52	VDD	Power		
53	PA4	I/O	DAC_OUT1	DAC_1
54	PA5	I/O	DAC_OUT2	DAC_2
57	PC4 *	I/O	GPIO_Input	GPIO5V_8
58	PC5 *	I/O	GPIO_Input	GPIO5V_7
59	VDD	Power		
60	VSS	Power		
61	PB0 *	I/O	GPIO_Input	GPIO5V_6

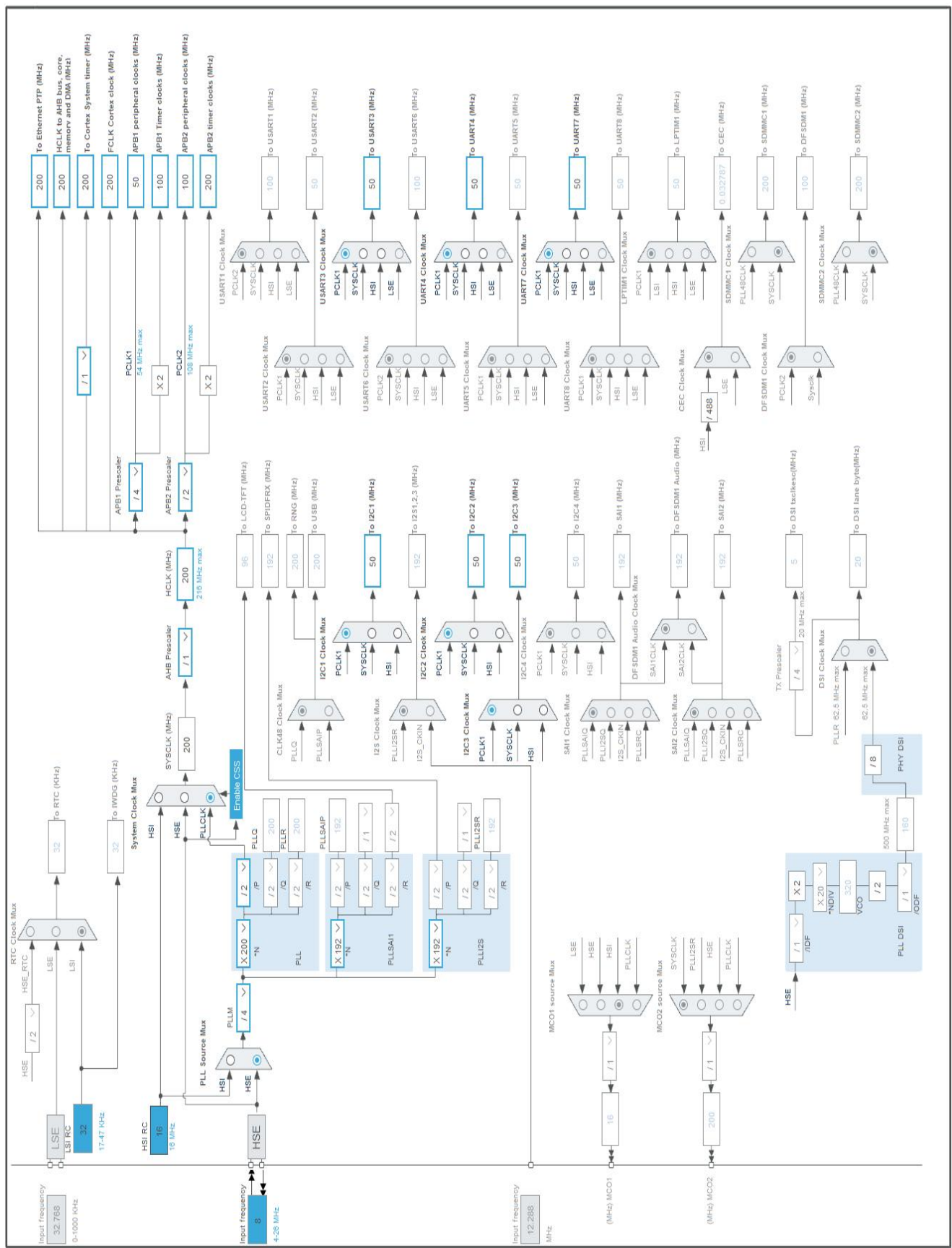
Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
62	PB1 *	I/O	GPIO_Input	GPIO5V_5
63	PB2 *	I/O	GPIO_Input	GPIO5V_4
64	PI15 *	I/O	GPIO_Input	GPIO5V_3
65	PJ0 *	I/O	GPIO_Output	GPIO5V_2
66	PJ1 *	I/O	GPIO_Output	GPIO5V_1
67	PJ2 *	I/O	GPIO_Output	GPIO5V_9
68	PJ3 *	I/O	GPIO_Output	GPIO5V_8
69	PJ4 *	I/O	GPIO_Output	GPIO5V_7
70	PF11 *	I/O	GPIO_Output	GPIO12V_6
71	PF12 *	I/O	GPIO_Output	GPIO12V_5
72	VSS	Power		
73	VDD	Power		
74	PF13 *	I/O	GPIO_Output	GPIO12V_4
75	PF14 *	I/O	GPIO_Output	GPIO12V_3
76	PF15 *	I/O	GPIO_Output	GPIO12V_2
77	PG0 *	I/O	GPIO_Output	GPIO12V_1
78	PG1 *	I/O	GPIO_Input	GPIO12V_10
79	PE7	I/O	UART7_RX	
80	PE8	I/O	UART7_TX	
81	PE9 *	I/O	GPIO_Input	GPIO12V_11
82	VSS	Power		
83	VDD	Power		
84	PE10 *	I/O	GPIO_Input	GPIO12V_12
85	PE11 *	I/O	GPIO_Input	GPIO12V_13
86	PE12 *	I/O	GPIO_Input	GPIO12V_14
87	PE13 *	I/O	GPIO_Input	GPIO12V_15
88	PE14 *	I/O	GPIO_Input	GPIO12V_16
89	PE15 *	I/O	GPIO_Input	GPIO12V_17
90	PB10	I/O	USART3_TX	
91	PB11	I/O	USART3_RX	
92	VCAP_1	Power		
93	VSS	Power		
94	VDD	Power		
97	PH7	I/O	I2C3_SCL	
98	PH8	I/O	I2C3_SDA	
99	PH9 *	I/O	GPIO_Input	GPIO12V_18
100	PH10 *	I/O	GPIO_Input	GPIO12V_19
101	PH11 *	I/O	GPIO_Input	GPIO12V_20
102	PH12 *	I/O	GPIO_Input	GPIO12V_21

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
103	VDD	Power		
104	PB12 *	I/O	GPIO_Output	GPIO3V_1
105	PB13 *	I/O	GPIO_Output	GPIO3V_2
106	PB14	I/O	GPIO_EXTI14	GPIO3V_3
107	PB15 *	I/O	GPIO_Output	GPIO3V_4
108	PD8 *	I/O	GPIO_Output	GPIO3V_5
109	PD9 *	I/O	GPIO_Output	GPIO3V_6
110	PD10 *	I/O	GPIO_Output	GPIO3V_7
111	PD11 *	I/O	GPIO_Output	GPIO3V_8
112	PD12 *	I/O	GPIO_Output	GPIO3V_9
113	PD13 *	I/O	GPIO_Output	GPIO3V_10
114	VSS	Power		
115	VDD	Power		
116	PD14 *	I/O	GPIO_Output	GPIO3V_11
117	PD15 *	I/O	GPIO_Output	GPIO3V_12
118	VDDDSI	Power		
119	VCAPDSI	Power		
122	VSSDSI	Power		
125	VDD12DSI	Power		
128	VSSDSI	Power		
136	VSS	Power		
137	VDDUSB	Power		
138	PC6	I/O	TIM8_CH1	PWM_8
139	PC7	I/O	TIM8_CH2	PWM_9
140	PC8	I/O	TIM8_CH3	PWM_10
141	PC9 *	I/O	GPIO_Input	GPIO3V_13
142	PA8	I/O	CAN3_RX	
143	PA9 *	I/O	GPIO_Input	GPIO5V_14
144	PA10 *	I/O	GPIO_Input	GPIO5V_15
145	PA11	I/O	CAN1_RX	
146	PA12	I/O	CAN1_TX	
147	PA13	I/O	SYS_JTMS-SWDIO	SWDIO
148	VCAP_2	Power		
149	VSS	Power		
150	VDD	Power		
158	VDD	Power		
159	PA14	I/O	SYS_JTCK-SWCLK	SWCLK
160	PA15	I/O	CAN3_TX	
170	VSS	Power		

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
171	VDDSDMMC	Power		
184	VSS	Power		
185	VDD	Power		
194	PB5	I/O	CAN2_RX	
195	PB6	I/O	CAN2_TX	
197	BOOT0	Boot		
198	PB8	I/O	I2C1_SCL	
199	PB9	I/O	I2C1_SDA	
202	VSS	Power		
203	PDR_ON	Reset		
204	VDD	Power		
208	PI7 *	I/O	GPIO_Output	LDAC_4

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F7
Line	STM32F7x9
MCU	STM32F769BITx
Datasheet	DS11532 Rev4

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

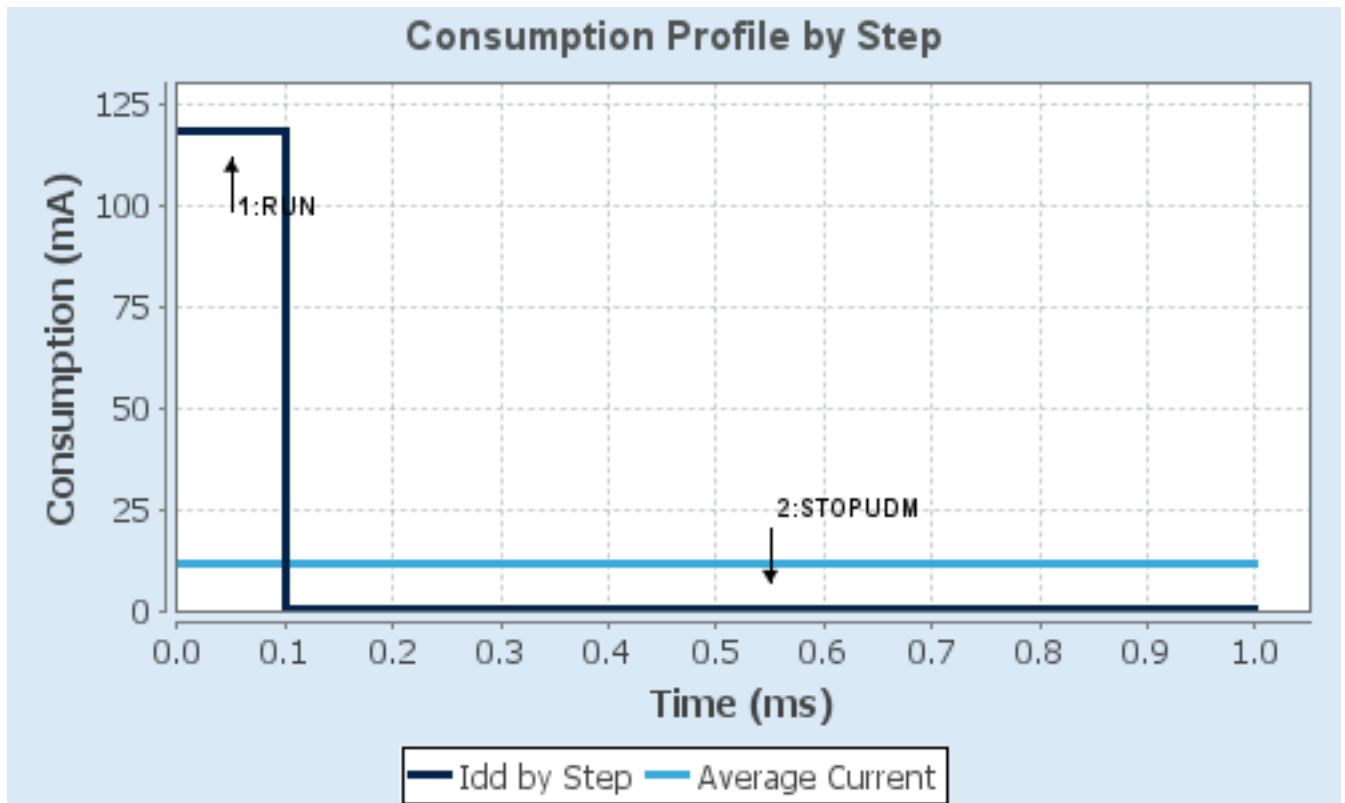
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP_UDM (Under Drive)
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	ICTM FLASH-SingleBank REGON	n/a
CPU Frequency	216 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	118 mA	130 μ A
Duration	0.1 ms	0.9 ms
DMIPS	462.0	0.0
Ta Max	97.6	104.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	11.92 mA
Battery Life	2 days, 4 hours	Average DMIPS	462.24005 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	HIL_2026
Project Folder	C:\Users\Joey\Documents\Coding Projects\UWFE\HIL\HIL_2026
Toolchain / IDE	Makefile
Firmware Package Name and Version	STM32Cube FW_F7 V1.17.4
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_CAN1_Init	CAN1
4	MX_CAN2_Init	CAN2
5	MX_CAN3_Init	CAN3
6	MX_DAC_Init	DAC
7	MX_I2C1_Init	I2C1
8	MX_I2C2_Init	I2C2
9	MX_I2C3_Init	I2C3
10	MX_SPI4_Init	SPI4
11	MX_SPI5_Init	SPI5

Rank	Function Name	Peripheral Instance Name
12	MX_TIM8_Init	TIM8
13	MX_UART4_Init	UART4
14	MX_UART7_Init	UART7
15	MX_USART3_UART_Init	USART3

3. Peripherals and Middlewares Configuration

3.1. CAN1

mode: Activated

3.1.1. Parameter Settings:

Bit Timings Parameters:

Prescaler (for Time Quantum)	5 *
Time Quantum	100.0 *
Time Quanta in Bit Segment 1	13 Times *
Time Quanta in Bit Segment 2	6 Times *
Time for one Bit	2000 *
Baud Rate	500000 *
ReSynchronization Jump Width	4 Times *

Basic Parameters:

Time Triggered Communication Mode	Disable
Automatic Bus-Off Management	Enable *
Automatic Wake-Up Mode	Enable *
Automatic Retransmission	Enable *
Receive Fifo Locked Mode	Disable
Transmit Fifo Priority	Disable

Advanced Parameters:

Operating Mode	Normal
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3.2. CAN2

mode: Activated

3.2.1. Parameter Settings:

Bit Timings Parameters:

Prescaler (for Time Quantum)	5 *
Time Quantum	100.0 *
Time Quanta in Bit Segment 1	13 Times *
Time Quanta in Bit Segment 2	6 Times *
Time for one Bit	2000 *
Baud Rate	500000 *
ReSynchronization Jump Width	4 Times *

Basic Parameters:

Time Triggered Communication Mode	Disable
Automatic Bus-Off Management	Enable *
Automatic Wake-Up Mode	Enable *
Automatic Retransmission	Enable *
Receive Fifo Locked Mode	Disable
Transmit Fifo Priority	Disable

Advanced Parameters:

Operating Mode	Normal
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3.3. CAN3

mode: Activated

3.3.1. Parameter Settings:

Bit Timings Parameters:

Prescaler (for Time Quantum)	10 *
Time Quantum	200.0 *
Time Quanta in Bit Segment 1	8 Times *
Time Quanta in Bit Segment 2	1 Time
Time for one Bit	2000 *
Baud Rate	500000 *
ReSynchronization Jump Width	1 Time

Basic Parameters:

Time Triggered Communication Mode	Disable
Automatic Bus-Off Management	Disable
Automatic Wake-Up Mode	Disable
Automatic Retransmission	Disable
Receive Fifo Locked Mode	Disable
Transmit Fifo Priority	Disable

Advanced Parameters:

Operating Mode	Normal
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3.4. CORTEX_M7

3.4.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode **Enabled ***

Cortex Interface Settings:

Flash Interface	AXI Interface
ART ACCELERATOR	Disabled
Instruction Prefetch	Disabled
CPU ICache	Disabled
CPU DCache	Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers
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Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	0x0 *
MPU Region Size	4GB
MPU SubRegion Disable	0x87 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	ENABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled
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3.5. DAC

mode: OUT1 Configuration

mode: OUT2 Configuration

3.5.1. Parameter Settings:

DAC Out1 Settings:

Output Buffer	Enable
Trigger	None

DAC Out2 Settings:

Output Buffer	Enable
Trigger	None

3.6. I2C1

I2C: I2C

3.6.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00C0EAFB *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.7. I2C2

I2C: I2C

3.7.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0

Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00C0EAF F *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.8. I2C3

I2C: I2C

3.8.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00C0EAF F *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.9.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	6 WS (7 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Over Drive	Enabled
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

3.10. SPI4

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.10.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	50.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware

3.11. SPI5

Mode: Full-Duplex Master

Hardware NSS Signal: Hardware NSS Output Signal

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	50.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Output Hardware

3.12. SYS

Debug: Serial Wire**Timebase Source: TIM1**

3.13. TIM8

Channel1: PWM Generation CH1**Channel2: PWM Generation CH2****Channel3: PWM Generation CH3****3.13.1. Parameter Settings:****Counter Settings:**

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.14. UART4

Mode: Asynchronous

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
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Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. UART7

Mode: Asynchronous

3.15.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.16. USART3

Mode: Asynchronous

3.16.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.17. FREERTOS

Interface: CMSIS_V1

3.17.1. Config parameters:

API:

FreeRTOS API	CMSIS v1
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Versions:

FreeRTOS version	10.2.1
CMSIS-RTOS version	1.02

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	7
MINIMAL_STACK_SIZE	256 *
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Disabled
USE_COUNTING_SEMAPHORES	Disabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Enabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	50000 *
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Option2 *

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Disabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Disabled
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Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

3.17.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled *
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Disabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled *
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Disabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Disabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.17.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
CAN1	PA11	CAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA12	CAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
CAN2	PB5	CAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB6	CAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
CAN3	PA8	CAN3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA15	CAN3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
DAC	PA4	DAC_OUT1	Analog mode	No pull-up and no pull-down	n/a	DAC_1
	PA5	DAC_OUT2	Analog mode	No pull-up and no pull-down	n/a	DAC_2
I2C1	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PB9	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
I2C2	PF0	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PF1	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
I2C3	PH7	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PH8	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
RCC	PH0/OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	OSC_OUT
	PH1/OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	OSC_IN
SPI4	PE2	SPI4_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PE4	SPI4_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PE5	SPI4_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PE6	SPI4_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SPI5	PF6	SPI5_NSS	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PF7	SPI5_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PF8	SPI5_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PF9	SPI5_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	SWCLK
TIM8	PC6	TIM8_CH1	Alternate Function Push Pull	No pull-up and no pull-down	High *	PWM_8
	PC7	TIM8_CH2	Alternate Function Push Pull	No pull-up and no pull-down	High *	PWM_9
	PC8	TIM8_CH3	Alternate Function Push Pull	No pull-up and no pull-down	High *	PWM_10
UART4	PA0/WKUP	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA1	UART4_RX	Alternate Function Push Pull	Pull-up *	Very High *	
UART7	PE7	UART7_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PE8	UART7_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
USART3	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB11	USART3_RX	Alternate Function Push Pull	Pull-up *	Very High *	
GPIO	PF10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LDAC_1
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LDAC_2
	PA3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LDAC_3
	PC4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_8
	PC5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_7
	PB0	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_6
	PB1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_5

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_4
	PI15	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_3
	PJ0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO5V_2
	PJ1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO5V_1
	PJ2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO5V_9
	PJ3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO5V_8
	PJ4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO5V_7
	PF11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO12V_6
	PF12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO12V_5
	PF13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO12V_4
	PF14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO12V_3
	PF15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO12V_2
	PG0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO12V_1
	PG1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_10
	PE9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_11
	PE10	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_12
	PE11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_13
	PE12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_14
	PE13	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_15
	PE14	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_16
	PE15	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_17
	PH9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_18
	PH10	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_19
	PH11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_20
	PH12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO12V_21
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_1
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_2
	PB14	GPIO_EXTI14	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	GPIO3V_3
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_4
	PD8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_5
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_6
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_7
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_8
	PD12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_9
	PD13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_10
	PD14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_11
	PD15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO3V_12
	PC9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO3V_13
	PA9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_14
	PA10	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO5V_15

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PI7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LDAC_4

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
TIM1 update interrupt and TIM10 global interrupt	true	15	0
CAN3 TX interrupt	true	5	0
CAN3 RX0 interrupt	true	5	0
CAN3 RX1 interrupt	true	5	0
CAN3 SCE interrupt	true	5	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
CAN1 TX interrupts	unused		
CAN1 RX0 interrupts	unused		
CAN1 RX1 interrupt	unused		
CAN1 SCE interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
USART3 global interrupt	unused		
EXTI line[15:10] interrupts	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 update interrupt and TIM13 global interrupt	unused		
TIM8 trigger and commutation interrupts and TIM14 global interrupt	unused		
TIM8 capture compare interrupt	unused		
UART4 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	unused		
CAN2 TX interrupts	unused		
CAN2 RX0 interrupts	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
CAN2 RX1 interrupt		unused	
CAN2 SCE interrupt		unused	
I2C3 event interrupt		unused	
I2C3 error interrupt		unused	
FPU global interrupt		unused	
UART7 global interrupt		unused	
SPI4 global interrupt		unused	
SPI5 global interrupt		unused	

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
TIM1 update interrupt and TIM10 global interrupt	false	true	true
CAN3 TX interrupt	false	true	true
CAN3 RX0 interrupt	false	true	true
CAN3 RX1 interrupt	false	true	true
CAN3 SCE interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f7_bsd.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32gui.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4031-using-the-stm32f2-stm32f4-and-stm32f7-series-dma-controller-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpdp-sink-application-with-stm32cubemx-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-

for related Tools & Software	and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmibus-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf
Design Notes & Tips	https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0334-stm32f76xxx-and-stm32f77xxx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00273119.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0410-stm32f76xxx-and-stm32f77xxx-advanced-armbased-32bit-mcus-stmicroelectronics.pdf

Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
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